$$
\begin{array}{l}
\qquad\qquad\qquad [k \in \mathit{Keys} \mapsto \\
\qquad\qquad\qquad\quad \textsc{if}\ \mathit{tx\_buffer}[t][k] \neq \langle\rangle \\
\qquad\qquad\qquad\quad \textsc{then}\ \mathit{store}[k] \cup \\
\qquad\qquad\qquad\qquad\quad \{[\mathit{val} \mapsto \mathit{tx\_buffer}[t][k][1], \\
\qquad\qquad\qquad\qquad\qquad \mathit{cts} \mapsto \mathit{shard\_ct}', \\
\qquad\qquad\qquad\qquad\qquad \mathit{writer} \mapsto t \\
\qquad\qquad\qquad\qquad\qquad] \\
\qquad\qquad\qquad\qquad\quad \} \\
\qquad\qquad\qquad\quad \textsc{else}\ \ \mathit{store}[k] \\
\qquad\qquad\qquad] \\
\qquad\qquad \land\ \mathit{client\_dep}' = \\
\qquad\qquad\qquad \textsc{if}\ \mathit{shard\_ct}' > \mathit{client\_dep}[\mathit{Workload}[t].\mathit{session}] \\
\qquad\qquad\qquad\ \ \textsc{then}\ [\mathit{client\_dep}\ \textsc{except}\ ![\mathit{Workload}[t].\mathit{session}] = \mathit{shard\_ct}'] \\
\qquad\qquad\qquad\ \ \textsc{else}\ \ \mathit{client\_dep} \\
\qquad\qquad \land\ \mathit{client\_last\_commit}' = \\
\qquad\qquad\qquad [\mathit{client\_last\_commit}\ \textsc{except} \\
\qquad\qquad\qquad ![\mathit{Workload}[t].\mathit{session}] = \\
\qquad\qquad\qquad \mathit{shard\_ct}'] \\
\qquad\qquad \land\ \textsc{unchanged}\ \langle \mathit{router\_ct}, \\
\qquad\qquad\qquad\quad \mathit{tx\_snapshot}, \\
\qquad\qquad\qquad\quad \mathit{tx\_checkset}, \\
\qquad\qquad\qquad\quad \mathit{tx\_buffer}, \\
\qquad\qquad\qquad\quad \mathit{pc}, \\
\qquad\qquad\qquad\quad \mathit{op\_seq} \\
\qquad\qquad\quad \rangle
\end{array}
$$

$$
\begin{array}{l}
\mathit{Next}\ \triangleq \\
\quad (\exists\, t \in \mathit{TxnIds} : \mathit{StartTx}(t) \lor \mathit{ExecuteOp}(t) \lor \mathit{CommitTx}(t)) \lor \\
\quad\ (\forall\, t \in \mathit{TxnIds} : \\
\qquad\quad \mathit{tx\_status}[t] \in \{\text{“Committed”}, \text{“Aborted”}\} \land \textsc{unchanged}\ \mathit{all\_vars} \\
\quad\ )
\end{array}
$$

2. AXIOMATIC FRAMEWORK & MAPPING

$$
\mathit{CommittedTxns}\ \triangleq\ \{t \in \mathit{AllTxns} : \mathit{tx\_status}[t] = \text{“Committed”}\}
$$

$$
\begin{array}{l}
\mathit{Derived\_AR}\ \triangleq \\
\quad \{\langle t1, t2\rangle \in \mathit{CommittedTxns} \times \mathit{CommittedTxns} : \\
\qquad \mathit{tx\_commit\_time}[t1] < \mathit{tx\_commit\_time}[t2] \\
\quad \}
\end{array}
$$

$$
\begin{array}{l}
\mathit{Derived\_VIS}\ \triangleq \\
\quad \{\langle t1, t2\rangle \in \mathit{CommittedTxns} \times \mathit{CommittedTxns} : \\
\qquad \lor\ \land\ t2 \neq \mathit{InitTx} \\
\qquad\quad\ \ \land\ \mathit{Workload}[t2].\mathit{level}\ \ \in \{\text{“RA”}, \text{“CC”}, \text{“PC”}, \text{“PSI”}, \text{“SI”}\}
\end{array}
$$